VeriCBAM: Master Reference & Source Verification Guide

Multimodal Evidence Fusion & Decision Support for EU Carbon Border Adjustment Mechanism (CBAM) Auditing

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Purpose: Authoritative Reference on Theory, EU Laws & Regulations, Participating Entities, Industrial Cohort (30 Plants), and System Architecture for **Google NotebookLM** Source Grounding | **Date:** October 2026 (v2.1)

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1. The Problem & How VeriCBAM Supports Field Professionals

On **1 January 2026**, the European Union's **Carbon Border Adjustment Mechanism (CBAM)** transitioned from its reporting-only phase (2023–2025) into its **definitive financial enforcement regime** under **Regulation (EU) 2023/956**. European importers (*Authorised CBAM Declarants*) of carbon-intensive commodities—**Cement, Iron & Steel, Aluminium, Fertilisers, Electricity, and Hydrogen**—must now surrender CBAM certificates priced at weekly EU ETS allowance auction rates (€65–€90+ per metric tonne of CO2) for the verified embedded emissions of their imports. While the **Omnibus I Simplification Regulation (EU) 2025/2083** introduced a 50-tonne annual mass *de minimis* exemption for small importers, approximately **99% of embedded industrial emissions** remain subject to mandatory verification and financial surrender.

1.1 Core Operational & Verification Bottlenecks

- **Severe Supply-Chain Information Asymmetry:** Unlike EU installations monitored for 20 years under the EU ETS Monitoring and Reporting Regulation (**Regulation (EU) 2018/2066**), CBAM regulates production at thousands of non-EU installations outside European regulatory jurisdiction, relying heavily on operator self-declarations.
- **Verification Capacity Constraints:** Under **Commission Implementing Regulation (EU) 2025/2546** and **Commission Delegated Regulation (EU) 2025/2551**, physical site visits are the regulatory default for initial verification, and verifiers must enforce a **5% materiality threshold at the goods (CN code) level**. However, accredited verifier capacity is globally constrained and physical overseas inspections of integrated steelworks or cement kilns are costly and time-intensive.
- **Three Technical Inconsistency Vectors:** (1) *Selective Quantitative Understatement* (under-reporting fuel/process CO2 by 10%–50%), (2) *Stoichiometric Floor & Boundary Manipulation* (claiming emissions below chemical conservation-of-mass limits or exporting blast furnace off-gases across legal fence lines without reconciliation), and (3) *Process Route Misclassification* (claiming low-carbon Scrap-EAF or DRI-EAF intensities for high-carbon primary BF-BOF steel).

1.2 How VeriCBAM Supports Specific Professionals in the Ecosystem

Professional User Group	Operational Challenge	How VeriCBAM Supports Their Daily Workflow
1. National Competent Authorities (NCAs) & Customs Officers (e.g., DEHSt, DGEC, NEa, KOBiZE, DG TAXUD)	Must screen thousands of annual CBAM declarations in the central registry with limited desk-audit staff.	Stage 1 Customs Triage (Model E2 Logistic Regression): Ranks incoming declarations by inconsistency risk (ROC-AUC = 0.9643). Auditing the top 20% highest-risk files yields 100.0% precision (2.71x lift over random selection) and catches 54.2% of all non-compliant declarations (75.0% recall at 30% budget).
2. Accredited CBAM Verifiers & Lead Auditors (e.g., TÜV SÜD, DNV, SGS, Bureau Veritas, LRQA, GUTcert)	Must perform strategic & risk analysis under Reg. (EU) 2025/2546 to justify site-visit waivers or focus on-site physical inspections.	Stage 2 Audit Dossier Engine (Model E1 Bayesian Fusion): Generates a 100% white-box, physically grounded Audit Dossier with near-zero false alarms (0.56% Authentic FPR; ROC-AUC = 0.8935). Lists exact stoichiometric margins, NOx/CO2 combustion drift, and specific documents to request on-site.
3. EU Authorised CBAM Declarants (Importers) & Non-EU Industrial Exporters (Commodity traders, steel/cement mills)	Face severe financial penalties (€100+/t CO2 under Art. 26 of Reg. 2023/956) and shipment rejection if supplier declarations are physically implausible.	Tab 4 Standalone Supplier Plausibility Pre-Check: Zero-history screening tool requiring only 4 inputs from a supplier communication sheet. Instantly verifies whether declared specific emissions violate chemical stoichiometry or indicate route misclassification prior to customs filing.

2. Complete Legal, Regulatory & Standard Framework (with EUR-Lex ELIs)

Every regulatory threshold, boundary rule, and verification workflow in VeriCBAM is anchored in primary European Union legislation and international greenhouse gas accounting standards:

Law / Standard	Short Title	Key Provisions & Role in VeriCBAM	Official ELI / Reference URL
Regulation (EU) 2023/956 (10 May 2023)	Primary CBAM Founding Regulation	Establishes CBAM scope (Annex I CN codes), embedded emissions methodology (Annex IV), declarant authorization (Art. 5), mandatory verification (Art. 8), accredited verifiers (Art. 18), and excess emissions penalties (Art. 26).	http://data.europa.eu/eli/reg/2023/956/oj
Implementing Regulation (EU) 2023/1773 (17 Aug 2023)	Transitional Period Reporting Rules	Defines installation boundaries, production routes, precursor accounting, and operator communication templates during the transitional phase (Oct 2023 – Dec 2025).	http://data.europa.eu/eli/reg_impl/2023/1773/oj
Implementing Regulation (EU) 2025/2546 (10 Dec 2025)	Definitive Verification Principles	Governs verification of declared embedded emissions from 2026 onward. Article 5 sets the 5% materiality threshold at the goods (CN code) level; sets physical site visits as the default and defines waiver criteria.	http://data.europa.eu/eli/reg_impl/2025/2546/oj
Implementing Regulation (EU) 2025/2547 (10 Dec 2025)	Calculation & Monitoring of Embedded Emissions	Defines system boundaries, functional units, precursor tracking (clinker, pig iron, DRI), default value mark-ups, and rules for transferred CO2 and waste gases (blast furnace gas, coke oven gas).	http://data.europa.eu/eli/reg_impl/2025/2547/oj
Delegated Regulation (EU) 2025/2551 (10 Dec 2025)	Accreditation & Oversight of CBAM Verifiers	Supplements Reg. 2023/956 with accreditation requirements, verifier technical competence criteria, National Accreditation Body (NAB) oversight, and mutual recognition.	http://data.europa.eu/eli/reg_del/2025/2551/oj
Regulation (EU) 2025/2083 (Omnibus I, 2025)	CBAM Simplification & De Minimis Threshold	Introduces a cumulative 50-tonne annual mass de minimis exemption for small importers while maintaining ~99% of embedded emissions within full verification.	http://data.europa.eu/eli/reg/2025/2083/oj
Implementing Reg. (EU) 2018/2066 & 2018/2067	EU ETS Monitoring & Reporting (MRR) & Verification (AVR)	Foundational EU ETS rules providing standard stoichiometric calculation tiers, mass-balance equations, and risk-based verification auditing practices.	http://data.europa.eu/eli/reg_impl/2018/2066/oj
Directive 2010/75/EU & Reg. (EC) 166/2006 / (EU) 2024/1244	Industrial Emissions Directive (IED) & E-PRTR / IEPR	Legal mandate requiring European heavy industrial facilities to report verified annual air releases (CO2, NOx, SOx), forming VeriCBAM's Layer A reference cohort.	http://data.europa.eu/eli/dir/2010/75/oj
Decisions 2013/163/EU & 2012/135/EU	EU BAT Conclusions (CLM BREF & Iron/Steel BREF)	JRC Best Available Techniques reference documents establishing physical thermal energy minimums for cement kilns (2.9–3.3 GJ/t) and blast furnace reducing agents.	http://data.europa.eu/eli/dec_impl/2013/163/oj
IPCC 2006 / 2019 Refinement (Vol. 3 IPPU) & ISO 14064-3	International GHG Stoichiometry & Audit Standards	Volume 3 Ch. 2 (Mineral/Cement calcination factor 0.7848 t CO2/t CaO) & Ch. 4 (Iron/Steel carbon balance); ISO 14064-3 verification standard; EN 14181 CEMS uncertainty.	https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol3.html

3. Core Scientific Theory & Mathematical Formulations

3.1 Chemical Engineering Stoichiometry (Conservation of Mass Bounds)

VeriCBAM evaluates whether declared Scope 1 emissions are physically feasible given the chemical reactions required to manufacture cement clinker or crude steel:

A. Cement Clinker Calcination & BAT Thermal Floor:
In a rotary kiln, limestone (CaCO3, 100.0869 g/mol) decomposes into quicklime (CaO, 56.0774 g/mol) and CO2 (44.0095 g/mol). By conservation of mass, the exact stoichiometric ratio is 44.0095 / 56.0774 = **0.7848 t CO2 per tonne of CaO**. Standard Portland clinker contains 63%–67% CaO (conservative minimum 63%), yielding a pure chemical calcination floor of **0.495 t CO2/t clinker** (central 0.525 t CO2/t). Combined with the Best Available Techniques (EU CLM BREF 2013) thermal kiln combustion minimum of **0.275 t CO2/t clinker** (for a 5-stage cyclone preheater/precalciner kiln requiring 2.9–3.3 GJ/t), the route-specific engineering lower bound is:

```
CaCO3 -> CaO + CO2 | E_floor,cement = P_clinker * (0.495 [calc] + 0.275 [BAT fuel]) - E_CCUS = 0.770 t CO2 / t clinker
```

B. Primary & Secondary Steelmaking Route Floors:
• **Integrated Blast Furnace – Basic Oxygen Furnace (BF-BOF):** Hematite reduction (Fe2O3 + 3C -> 2Fe + 3CO / 3CO2) requires a theoretical minimum of 1.182 t CO2/t hot metal. Including blast furnace thermal enthalpy balance, sintering/coking, and up to 15%–20% scrap dilution in the BOF converter yields a route-specific engineering lower bound of **1.350 t CO2/t crude steel**.
• **Direct Reduced Iron – Electric Arc Furnace (DRI-EAF):** Natural gas shaft reduction yields a direct Scope 1 lower bound of **0.550 t CO2/t crude steel**.
• **Secondary Scrap – Electric Arc Furnace (Scrap-EAF):** Melting 100% recycled ferrous scrap requires no ore reduction; direct Scope 1 emissions from graphite electrodes and burners have a lower bound of **0.040 t CO2/t crude steel**.

C. Conservative Utilization Proxy & Transferred Process Gas Boundary Rule (Step 7 Resolution):
When auditing against external registry nameplate capacity (rather than verified production), VeriCBAM evaluates the conservative floor at minimum plausible utilization (**50% of nameplate capacity**: prod_min = 0.50 * Capacity). Furthermore, in 5 major European integrated steelworks (*ArcelorMittal Bremen, Eisenhüttenstadt, Taranto, HKM Duisburg, Tata Steel IJmuiden*), carbon-rich blast furnace gas (BFG) is exported across the legal fence line to adjacent utility power stations operating under separate E-PRTR IDs. VeriCBAM flags these boundary-mismatched plants (stoich_applicable = False) so the stoichiometry stream explicitly abstains (NaN), eliminating all 34 historical false-positive floor breaches (**0 authentic breaches across 177 real facility-years**).

3.2 Multi-Pollutant Combustion Consistency (NOx/CO2) & Longitudinal Trajectory

• **Leave-One-Year-Out NOx/CO2 Ratio:** High-temperature combustion co-emits NOx and CO2. Across 28 European plants, the within-facility annual NOx/CO2 ratio has a median Coefficient of Variation of **11.1%**. If declared CO2 drops sharply while reported NOx remains high, the leave-one-year-out Z-score flags selective CO2 understatement.
• **One-Sided Pooled MAD Historical Score:** To avoid penalizing facilities with short histories or legitimate output growth, longitudinal deviations are scaled by the cohort-wide pooled Median Absolute Deviation (**sigma_pooled = 0.0911** across 177 plant-years) using a strictly one-sided drop score:

```
z_drop = max( 0, - [ ln(E_declared) - Median_hist(ln E_CO2) ] / sigma_pooled ) where sigma_pooled = 0.0911
```

3.3 Copernicus Sentinel-5P Satellite Theory & Documented Physical Limitations

VeriCBAM extracts authentic Level-2 tropospheric NO2 vertical column density granules (S5P_L2__NO2____, qa_value >= 0.50) via Microsoft Planetary Computer STAC and CDSE, computing spatial plume contrast between a 5 km plant radius and a 15–30 km background ring. **Crucial Scientific Finding:** Empirical validation across 28 facilities (cross-sectional Spearman rho = 0.198, p = 0.312) and multi-year 2019–2023 CDSE time series (mean within-plant rho = -0.100) confirms that 5.5 x 3.5 km orbital NO2 columns cannot quantify annual CO2 due to cloud obscuration, wind advection, and variable NOx chemistry. Satellite is therefore restricted to an auxiliary qualitative check (w_sat = 1.0) with cloud confidence gating C_sat = max(0, 1 - cloud_fraction).

3.4 Multimodal Fusion Equation (Model E1) vs. Supervised Triage (Model E2)

```
logit P(H | E) = logit P(H) + SUM_{k=1..6} [ w_k * C_k * ln(LR_k) ] with Prior P(H) = 0.08 and Gate C_overall >= 0.50
```

• **Model E1 (Expert Weighted Log-Odds Bayesian Fusion):** Combines Route Match (w=2.0), Stoichiometry (w=1.5), NOx/CO2 Ratio (w=1.2), Historical Drop (w=1.0), Satellite Plume (w=1.0), and Registry Utilization (w=0.8) with 500-draw Monte Carlo 95% credible intervals and an automatic *Insufficient Evidence* gate when C_overall < 0.50.
• **Model E2 (Supervised Logistic Regression):** Trained via **5-Fold GroupKFold Cross-Validation** grouped strictly by facility_id (zero facility leakage between folds) to maximize customs triage ranking.

4. Participating Institutions, Regulatory Agencies & Data Bodies

Institution / Entity Name	Location	Role in the CBAM & VeriCBAM Ecosystem
European Commission — DG TAXUD & DG CLIMA	Brussels, EU	Central CBAM Registry operator, declarant authorization, EU ETS price alignment, and enforcement policy.
Joint Research Centre — European IPPC Bureau (JRC)	Seville, Spain	Authors CBAM technical embedded emissions methodology and official EU BAT Reference Documents (Cement & Steel BREFs).
European Environment Agency (EEA)	Copenhagen, EU	Publishes the E-PRTR / IED v16 Industrial Reporting Database (DOI: 10.2909/657ac3cb-affa-4295-a4a9-27b4f539adab).
European Space Agency (ESA) & Copernicus / CDSE	Paris / EU	Operates Sentinel-5P TROPOMI satellite and Copernicus Data Space Ecosystem API (paired with Microsoft Planetary Computer STAC).
Global Energy Monitor (GEM)	International	Maintains the Global Steel Plant Tracker & Global Cement and Concrete Tracker providing independent plant capacities and routes.
National Competent Authorities (NCAs) & Env. Agencies	EU Member States	DEHSt/UBA (Germany), GIOŚ/KOBiZE (Poland), UBA-AT (Austria), DGEC/GEREP (France), NEa/RIVM (Netherlands), ISPRA (Italy), NRZ (Slovakia), Miljøstyrelsen (Denmark), Naturvårdsverket (Sweden), Ministry of Env. (Greece).
Industry Technical Associations	Brussels / Düsseldorf	CEMBUREAU (European Cement Assoc.), VDZ (German Cement Works Assoc.), EUROFER (European Steel Assoc.), worldsteel.
Accreditation Bodies (NABs) & Verification Bodies (VBs)	Global / EU	NABs: DAkkS (DE), COFRAC (FR), RvA (NL), PCA (PL), ACCREDIA (IT). Accredited Verifiers: TÜV SÜD, DNV, SGS, Bureau Veritas, LRQA, GUTcert, RINA.

5. Complete Directory of the 30 Participating Industrial Companies & Plants

The VeriCBAM empirical benchmark is built on 177 authentic reporting years (2018–2023) across 30 major European installations (15 Cement plants and 15 Iron & Steel plants), cross-referenced against Global Energy Monitor, national permits, and corporate filings:

5.1 Cement Sector Installations (15 Plants)

Facility Name (E-PRTR)	Parent Company	City, Country	Verified Process Route	Clinker/Cem (Mtpa)	Util.	Primary Source
1. CEMEX Zement GmbH	CEMEX S.A.B. de C.V.	Rüdersdorf, DE	Dry Kiln (4-Stage Precalciner)	2.00 / 2.40	80%	cemex.de
2. Dyckerhoff GmbH, Werk Deuna	Buzzi S.p.A. (Dyckerhoff)	Niederorschel, DE	Dry Kiln (Multi-Stage Precalciner)	1.50 / 1.80	78%	dyckerhoff.com
3. Dyckerhoff GmbH (Lengerich)	Buzzi S.p.A. (Dyckerhoff)	Lengerich, DE	Dry Kiln (Preheater/Precalciner)	1.20 / 1.50	76%	dyckerhoff.com
4. Holcim (Deutschland) GmbH	Holcim Group	Lägerdorf, DE	Semi-Dry / Preheater (Oxyfuel Project)	1.20 / 1.40	80%	holcim.de
5. Heidelberg Materials (Burglengenfeld)	Heidelberg Materials AG	Burglengenfeld, DE	Dry Kiln (Grate Cooler & Precalciner)	1.10 / 1.30	78%	heidelbergmaterials.de
6. HeidelbergCement (Schelklingen)	Heidelberg Materials AG	Schelklingen, DE	5-Stage Preheater Kiln (SCR de-NOx)	1.20 / 1.45	82%	heidelbergmaterials.de
7. Schwenk Zement (Allmendingen)	SCHWENK Zement KG	Allmendingen, DE	Dry Kiln (Oxyfuel Testing)	1.05 / 1.25	78%	schwenk.de
8. Zementwerk Rohrdorf	Rohrdorfer Gruppe	Rohrdorf, DE	Dry Kiln + CO2 Capture Pilot Line	1.15 / 1.40	80%	rohrdorfer.eu
9. Heidelberg Materials (Lengfurt)	Heidelberg Materials AG	Triefenstein, DE	Dry Kiln with Preheater/Precalciner	1.00 / 1.20	78%	heidelbergmaterials.de
10. Dyckerhoff GmbH (Göllheim)	Buzzi S.p.A. (Dyckerhoff)	Göllheim, DE	Dry Kiln with Preheater/Precalciner	0.95 / 1.20	76%	dyckerhoff.com
11. Górażdże Cement S.A.	Heidelberg Materials AG	Chorula, PL	Dry Kiln (Largest in Central Europe)	3.60 / 4.20	85%	gorazdze.pl
12. HERACLES G.C.Co, Milaki Plant	Holcim Group (Heracles)	Aliveri, GR	Dry Kiln with Precalciner (Export)	2.20 / 2.60	85%	lafarge.gr
13. Grupa Ożarów S.A.	CRH plc	Karsy, PL	Dry Kiln with Preheater/Precalciner	2.40 / 2.90	80%	ozarow.com.pl
14. Aalborg Portland A/S	Cementir Holding N.V.	Aalborg, DK	Dry + Semi-Wet White/Grey Kilns	2.50 / 3.20	85%	aalborgportland.dk
15. Heidelberg Materials Slitefabriken	Heidelberg Materials AG	Slite, SE	Dry Kiln (Gotland Full-Scale CCS)	2.00 / 2.30	82%	heidelbergmaterials.se

5.2 Iron & Steel Sector Installations (15 Plants)

Facility Name (E-PRTR)	Parent Company	City, Country	Verified Process Route & Boundary	Steel/Iron (Mtpa)	Util.	Primary Source
16. Salzgitter Flachstahl GmbH	Salzgitter AG	Salzgitter, DE	BF-BOF Integrated (3 BFs, 3 BOFs)	4.70 / 4.40	86%	salzgitter-ag.com
17. Hüttenwerke Krupp Mannesmann	tk / Salzgitter / Vallourec	Duisburg, DE	BF-BOF Integrated [Transferred BFG]	5.60 / 5.20	45%	hkm.de
18. ArcelorMittal Bremen GmbH	ArcelorMittal S.A.	Bremen, DE	BF-BOF Integrated [Transferred BFG]	3.60 / 3.40	38%	bremen.arcelormittal.com
19. ArcelorMittal Eisenhüttenstadt	ArcelorMittal S.A.	Eisenhüttenstadt, DE	BF-BOF Integrated [Transferred BFG]	2.40 / 2.00	42%	eisenhuettenstadt.arcelormittal.com
20. thyssenkrupp Werk Hamborn	thyssenkrupp Steel AG	Duisburg, DE	Partial Site: Schwelgern BFs Only	0.00 / 7.20	12%	thyssenkrupp-steel.com

Facility Name (E-PRTR)	Parent Company	City, Country	Verified Process Route & Boundary	Steel/Iron (Mtpa)	Util.	Primary Source
21. thyssenkrupp Werk Beeckerwerth	thyssenkrupp Steel AG	Duisburg, DE	Partial Site: Rolling & Coating Only	0.00 / 0.00	80%	thyssenkrupp-steel.com
22. DK Recycling und Roheisen	DK Recycling Group	Duisburg, DE	Ferrous Waste Blast Furnace (Pig Iron)	0.00 / 0.35	80%	dk-duisburg.de
23. thyssenkrupp Werk Bruckhausen	thyssenkrupp Steel AG	Duisburg, DE	Partial Site: BOF Converter Shop Only	5.80 / 0.00	80%	thyssenkrupp-steel.com
24. ArcelorMittal Hochfeld GmbH	ArcelorMittal S.A.	Duisburg, DE	Secondary Scrap-EAF Mini-Mill	1.05 / 0.00	82%	germany.arcelormittal.com
25. Georgsmarienhütte GmbH	GMH Gruppe	Georgsmarienhütte, DE	Secondary Scrap-EAF (DC Furnace)	1.00 / 0.00	75%	gmh-gruppe.de
26. voestalpine Stahl GmbH	voestalpine AG	Linz, AT	BF-BOF Integrated (3 Blast Furnaces)	6.00 / 5.40	78%	voestalpine.com
27. ArcelorMittal Méditerranée	ArcelorMittal S.A.	Fos-sur-Mer, FR	BF-BOF Integrated Coastal Works	4.50 / 4.20	80%	mediterranee.arcelormittal.com
28. Tata Steel IJmuiden BV	Tata Steel Ltd.	Velsen-Noord, NL	BF-BOF Integrated [Transferred BFG]	7.50 / 7.00	45%	tatasteel.nl
29. U.S. Steel Košice s.r.o.	U.S. Steel Corp.	Košice, SK	BF-BOF Integrated (3 Blast Furnaces)	4.50 / 4.20	78%	usske.sk
30. Stabilimento di Taranto	Acciaierie d'Italia (Ilva)	Taranto, IT	BF-BOF Integrated [Transferred BFG]	8.00 / 7.50	35%	gruppoadi.com

6. How VeriCBAM Works: Architecture, UI Cockpit & Empirical Results

VeriCBAM operates through an 8-Tab interactive Streamlit Decision-Support Cockpit backed by a 100% deterministic Python pipeline (reproduce_all.py, verified by 28/28 passing pytest unit and integration tests):

- **Tab 1 (Executive Overview):** System architecture, regulatory framing, and live benchmark KPIs.
- **Tab 2 (Facility Cohort & Geospatial Inspector):** Interactive map highlighting selected installations alongside longitudinal 2018–2023 CO2 timelines and legal threshold overlays.
- **Tab 3 (Stoichiometric Feasibility & Route Sandbox):** Chemical mass-balance calculator testing limestone calcination, iron reduction, scrap dilution, and CCUS deductions.
- **Tab 4 (Standalone Supplier Plausibility Pre-Check):** Zero-history screening tool for non-EU supplier declarations.
- **Tab 5 (Copernicus Sentinel-5P Satellite Inspector):** Orbital NO2 column density, background ring contrast, and cloud gating.
- **Tab 6 (Bayesian Evidence Fusion & Audit Dossier):** Log-odds waterfall chart, prior sensitivity analysis (P(H) in [2%, 15%]), and 1-click export of an ISO 14064-3 / Reg. 2025/2546 Verifier Audit Dossier.
- **Tab 7 (Empirical Benchmark Evaluation & Ablation):** Complete 6-model evaluation across 325 graded scenarios.
- **Tab 8 (Proposal, References & Roadmap):** Master academic metadata and direct EUR-Lex ELI links.

6.1 Definitive Empirical Benchmark Comparison (325 Graded Cases, Primary delta >= 10%)

Model Architecture	ROC-AUC (95% CI)	PR-AUC (95% CI)	Brier	ECE	Auth FPR	Rec@5% FPR	Rec@10% FPR	Abstain	Deployment Role
Model A: Naive Prior (P=0.08)	0.5000 [0.500, 0.500]	0.3692 [0.369, 0.369]	0.2329	0.3192	0.00%	0.00%	0.00%	0.0%	Null Baseline
Model B: Stoichiometry Alone	0.7909 [0.719, 0.891]	0.7326 [0.641, 0.845]	0.1884	0.1571	1.55%	33.73%	54.22%	28.6%	Hard Physical Veto
Model C: Satellite S-5P Alone	0.5183 [0.499, 0.544]	0.3781 [0.359, 0.396]	0.3731	0.3474	96.61%	6.67%	10.83%	0.0%	Auxiliary Check
Model D: Historical Alone	0.9631 [0.957, 0.972]	0.9428 [0.931, 0.957]	0.0826	0.0632	6.78%	83.33%	90.83%	0.0%	Historical Drift
Model E1: Expert Bayesian	0.8935 [0.865, 0.924]	0.8684 [0.832, 0.901]	0.1530	0.1652	0.56%	69.17%	70.00%	0.0%	Stage 2 Audit Dossier
Model E2: Learned Logistic	0.9643 [0.956, 0.973]	0.9484 [0.938, 0.961]	0.0787	0.0630	6.78%	85.00%	91.67%	0.0%	Stage 1 Port Triage

Key Methodological Findings & Caveats:

1. **Matched FPR Parity (Step 6):** At matched false-alarm rates (FPR = 5%, 10%), Model D (Historical Alone: 83.3% / 90.8% recall) performs on par with Model E2 (Learned Logistic: 85.0% / 91.7% recall). Multimodal Bayesian fusion (Model E1) serves a distinct legal purpose: physical sanity-checking, boundary-aware constraint verification, and explainable audit dossiers with near-zero false alarms (0.56% FPR).
2. **Biomass Data Sparsity (Step 11):** In the EEA E-PRTR database, 'CO2 excluding biomass' is missing (NaN) for 163 of 177 rows (92.1%); total direct CO2 is therefore the only complete pan-European regulatory series.
3. **Benchmark Prevalence Caveat:** The 325-case benchmark has a 36.92% positive class prevalence (120/325) across 30 European facilities; real-world customs prevalence is lower, and third-country generalization remains a stated research boundary.

7. Direct Primary Source URLs for Google NotebookLM Ingestion

Copy and paste these authoritative URLs directly into Google NotebookLM ('Add Source -> Website') to verify every law, dataset, and technical parameter cited in this project:

Source Title / Document Name	Direct Official URL for NotebookLM
1. Primary CBAM Regulation (EU) 2023/956	https://eur-lex.europa.eu/eli/reg/2023/956/oj
2. CBAM Transitional Reporting Reg. (EU) 2023/1773	https://eur-lex.europa.eu/eli/reg_impl/2023/1773/oj
3. CBAM Verification Principles Reg. (EU) 2025/2546	https://eur-lex.europa.eu/eli/reg_impl/2025/2546/oj

Source Title / Document Name	Direct Official URL for NotebookLM
4. CBAM Embedded Emissions Calculation Reg. (EU) 2025/2547	https://eur-lex.europa.eu/eli/reg_impl/2025/2547/oj
5. CBAM Verifier Accreditation Delegated Reg. (EU) 2025/2551	https://eur-lex.europa.eu/eli/reg_del/2025/2551/oj
6. EC DG TAXUD Official CBAM Portal	https://taxation-customs.ec.europa.eu/carbon-border-adjustment-mechanism_en
7. EEA European Industrial Emissions Portal (E-PRTR v16)	https://industry.eea.europa.eu/
8. EEA E-PRTR Dataset Catalogue Record (DOI Metadata)	https://sdi.eea.europa.eu/catalogue/srv/api/records/657ac3cb-affa-4295-a4a9-27b4f539adab
9. Global Energy Monitor — Global Steel Plant Tracker	https://globalenergymonitor.org/projects/global-steel-plant-tracker/
10. Global Energy Monitor — Global Cement and Concrete Tracker	https://globalenergymonitor.org/projects/global-cement-and-concrete-tracker/
11. IPCC 2019 Refinement to 2006 GHG Guidelines (Vol. 3 IPPU)	https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol3.html
12. EU JRC BAT Reference (BREF) — Cement & Lime (2013/163/EU)	https://eippcb.jrc.ec.europa.eu/reference/production-cement-lime-and-magnesium-oxide
13. EU JRC BAT Reference (BREF) — Iron & Steel (2012/135/EU)	https://eippcb.jrc.ec.europa.eu/reference/iron-and-steel-production
14. ESA Copernicus Sentinel-5P TROPOMI Mission Portal	https://sentiwiki.copernicus.eu/web/s5p-mission
15. VeriCBAM Open-Source GitHub Repository	https://github.com/CaceresCristhian/VeriCBAM